
Homework 1

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MA 564 Section A1

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Problem 1 [COMPLETE]

By definition, $\emptyset, \mathcal{T} \in \mathbb{R}$. Now let $\{(-\infty, p)\}_{p \in P} \subseteq \mathcal{T}$, and consider the union over this collection. Let $a = \sup P$, and let $x \in (-\infty, a)$. We have that $a < \sup P$, and thus by definition, there exists $p \in P$ such that $a < p$. It follows that $x \in (-\infty, p)$, and thus we have

$$x \in (-\infty, p) \subseteq \bigcup_{p \in P} (-\infty, p).$$

Therefore, $(-\infty, a) \subseteq \bigcup_{p \in P} (-\infty, p)$. Now let $x \in \bigcup_{p \in P} (-\infty, p)$. It follows that there exists $p \in P$ such that $x \in (-\infty, p)$, and thus $x < p$ for some $p \in P$. Since $p < \sup P = a$, we have that $x < a$, and thus $x \in (-\infty, a)$. Therefore,

$$\bigcup_{p \in P} (-\infty, p) = (-\infty, \sup P).$$

Since $(-\infty, \sup P)$ is of the same form as given in the definition of $\mathcal{T}$, we have $(-\infty, \sup P) \in \mathcal{T}$. Therefore, $\mathcal{T}$ is closed under union.

Now let P be a finite subset of $\mathbb{R}$, and consider the collection $\{(-\infty, p)\}_{p \in P}$. Since P is finite, it must have a least element. Let $a = \min P$ be this least element, and note that for all $p \in P$, $a \leq p$ implies that $(-\infty, a) \subseteq (-\infty, p)$. It follows that

$$(-\infty, a) \bigcap_{p \in P} (-\infty, p).$$

However, we also know that due to how intersections work,

$$\bigcap_{p \in P} (-\infty, p) \subseteq (-\infty, a).$$

Therefore, the finite intersection is of the form $(-\infty, a)$, and thus is an element of $\mathcal{T}$. We have that $\mathcal{T}$ is closed under finite intersection, and is thus a topology on $\mathbb{R}$.

Problem 2 [COMPLETE]

Let $n \in \mathbb{Z}$. If n is odd, we have $n \in \{n\} = B(n) \in \mathcal{B}$. If n is even, then $n \in \{n-1, n, n+1\} = B(n) \in \mathcal{B}$. Therefore, for all $n \in \mathbb{Z}$, there exists $B(n) \in \mathcal{B}$ such that $n \in B(n)$.

Now consider the intersection $B(n) \cap B(m)$ for $n, m \in \mathbb{Z}$. If $B(n) \cap B(m) = \emptyset$, then it is vacuously true that for all $x \in B(n) \cap B(m)$, there exists $B \in \mathcal{B}$ such that $x \in B \subseteq B(n) \cap B(m)$. Thus, assume without loss of generality that the intersection is nonempty.

Consider the case where either n or m is odd. Without loss of generality, assume n is odd. We thus have

$$n \in B(n) \cap B(m) \implies B(n) = \{n\} \subseteq B(n) \cap B(m).$$

Now consider the case where both n and m are even. If $n = m$, we trivially have $B(n) \subseteq B(n) \cap B(m)$ once again, since $B(n) = B(m)$, so assume without loss of generality that $n < m$. We then have

$$B(n) \cap B(m) = \{n-1, n, n+1\} \cap \{m-1, m, m+1\}.$$

Since $m > n$, but the intersection is nonempty, we must have $m-1 = n+1$, since if m is any greater the intersection is nonempty, but if m is any less than $m = n$ or $m < n$. Note that since n is even, $n+1$ is odd, so

$$B(n+1) \subseteq B(n) \cap B(m).$$

In all of these cases, note that $B(n), B(n+1) \in \mathcal{B}$, so we have that $\mathcal{B}$ is a basis for a topology on $\mathbb{Z}$.

Problem 3 [COMPLETE]

Note that $X^c = X \setminus X = \emptyset$, and $\emptyset^c = X \setminus \emptyset = X$. Therefore, $\emptyset, X$ are closed.

Let $\{X_a\}_{a \in A}$ be a collection of closed sets in X . It follows that there exists a collection $\{Y_a\}_{a \in A}$ of open sets such that $Y_a^c = X_a$. It follows that

$$\bigcap_{a \in A} X_a = \bigcap_{a \in A} Y_a^c.$$

By De Morgan's laws, this equals

$$\left(\bigcup_{a \in A} Y_a \right)^c.$$

By the definition of a topology, $\bigcup_{a \in A} Y_a$ is open, so $\left(\bigcup_{a \in A} Y_a \right)^c$ is closed.

Let $\{X_k\}_{k=1}^k$ be a finite collection of closed sets in X . It follows that there exists a collection $\{Y_i\}_{i=1}^k$ of open sets such that $Y_i^c = X_i$ for all i . By De Morgan's laws, it follows that

$$\bigcup_{i=1}^k X_i = \bigcup_{i=1}^k Y_i^c = \left(\bigcap_{i=1}^k Y_i \right)^c.$$

By definition of a topology, since Y_i is open for all i , it follows that $\bigcap_{i=1}^k Y_i$ is open. Therefore, $\left(\bigcap_{i=1}^k Y_i \right)^c$ is closed.

Problem 4

Problem 5

Let X be Hausdorff, and consider the intersection

$$U := \bigcap_V \overline{V},$$

where V ranges over all open sets in X containing x . By definition, we have $x \in \overline{V}$ for all V , so $x \in U$. Note that for all $y \in X$ such that $y \neq x$, since X is Hausdorff, there exist disjoint neighborhoods V_x, V_y of x, y , respectively. Since $U \subseteq V_x \subseteq \overline{V_x}$ and $y \notin V_x$, we have $y \notin U$ for all $y \neq x$ in X . Therefore, $U = \{x\}$.

Now for the converse, suppose $U = \{x\}$. Let $y \in X$ be a distinct point. Since $y \notin U$, it follows that there must exist at least one neighborhood V of x such that $y \notin \bar{V}$. Equivalently, $y \in \bar{V}^c$. Note that $\bar{V} \cap \bar{V}^c = \emptyset$. Since $V \subseteq \bar{V}$, we have $V \cap \bar{V}^c = \emptyset$. Since $\bar{V}$ is closed, $\bar{V}^c$ is open, and contains y . V is also open, and contains x . Therefore, for any two distinct points, there exist disjoint neighborhoods of those points, and X is Hausdorff.

Problem 6 [COMPLETE]

Let definitions be as given in the problem, and suppose X is Hausdorff. Then for all $x \neq y$ in X , there exist disjoint neighborhoods V_x, V_y of x and y , respectively. Equivalently, $V_x \subseteq X \setminus V_y = V_y^c$. Since V_y is open, V_y^c is finite, and thus V_x is finite. Note that $X = V_x \cup V_x^c$. Since V_x and V_x^c are finite, X is finite.

Now suppose X is finite. It follows that all subsets $U \subseteq X$ are finite, and thus $U^c = X \setminus U$ is finite. Therefore, $\mathcal{T} = \mathcal{P}(X)$ is the power set of X , and we have that for any points $x \neq y$ in X , the sets $\{x\}, \{y\} \in \mathcal{T}$ are trivially disjoint.

Problem 7 [COMPLETE]

Note that for any $C \in \mathcal{C}$, the intersection $\bigcap_{i=1}^1 C = C \in \mathcal{B}_{\mathcal{C}}$. Therefore, $\mathcal{C} \subseteq \mathcal{B}_{\mathcal{C}}$. By definition,

$$\bigcup_{C \in \mathcal{C}} C = X.$$

Therefore, for all $x \in X$, there exists $C \in \mathcal{B}_{\mathcal{C}}$ such that $x \in C$. Now let $U_1, U_2 \in \mathcal{B}_{\mathcal{C}}$. It follows that there exist finite collections $\{C_i\}_{i=1}^k, \{C_j\}_{j=1}^\ell \subseteq \mathcal{C}$ such that

$$U_1 = \bigcap_{i=1}^k C_i, \quad U_2 = \bigcap_{j=1}^\ell C_j.$$

Let $\mathcal{V}$ be the collection of all C_i s and C_j s. We have

$$U_1 \cap U_2 = \left(\bigcap_{i=1}^k C_i \right) \cap \left(\bigcap_{j=1}^\ell C_j \right) = \bigcap_{V \in \mathcal{V}} V.$$

Since $\{C_i\}_{i=1}^k$ and $\{C_j\}_{j=1}^\ell$ are finite, $\mathcal{V}$ is finite, and thus $U_1 \cap U_2$ is a finite intersection of elements of $\mathcal{C}$. Therefore, $U_1 \cap U_2 \in \mathcal{B}_{\mathcal{C}}$. Note that

$$U_1 \cap U_2 \subseteq U_1 \cap U_2,$$

and thus $\mathcal{B}_{\mathcal{C}}$ is a basis.